Weather Analyser üöä

General

Topic

The title of the software project is: Weather Analyser üöä

The software belongs to the research fields:

- Natural Sciences / Atmospheric Science and Oceanography
- Natural Sciences / Water Research

It is categorized in Application Class 3: For the software it is essential to avoid errors and to reduce risks. This applies in particular to critical software and that with product characteristics.

Its intended use and its contribution to research:

Intended use:
1. To bundle all pipeline steps from the raw data from Automatic Weather Stations (AWS) to a final dashboard,
2. To allow for the download of the data in different formats

Contribution to research: the software simplifies the quality control, and prepares them for publication

Its novelty:

It provides an easy-to-use interface that does not require any software background, and focuses in our specific use case

Contributor

- David Walter (https://orcid.org/0000-0001-6807-5007), Research Data Management at Max Planck Digital Library

Funding

Existing (financial or personnel) resources or specific funding for the software development:

DFG grant application

Schedule

Project's start date: May 1, 2025

Project's end date: April 30, 2027

Project Management

Development process and roles:

We will decide the project management method as a first project step

Tasks and use cases will be tracked with:

- OpenProject
- GitHub

There will not be any specification document outlining the software's most important requirements.

Development Requirements

Institutional requirements for software development:

We will follow https://hdl.handle.net/21.11116/0000-000C-80B1-A.

Requirements for software development from other parties:

No

Technical

Code

Programming language: Python

Technology or process used for versioning:

Third-Party Components and Libraries

Dependencies and documentation of used third-party software components and libraries:

We will use ZZZ, which is a software package for professional weather visualisations

Licenses of used third-party software components:

MIT

Tracking of external software components and dependency mitigation/elimination:

We will dockerized the project, so that we always work with a stable bundle of package versions

Planned use of third-party web services:

No

Reference to other software or objects:

No

Infrastructure

Needed infrastructure resources:

File server, deployment server, Git systems

Existing infrastructure for the software development:

Instute insfrastructure: file server, and deployment server.
Open source: Git versioning, and GitHub

Needed technical training or code-reviewing process:

Python course

Preservation

Time horizon and taken measures for software reusability:

At least for the project's duration will the software remain usable and maintained. However, it is expected that our scientific processing group will further develop and maintain the software also afterwards

Time horizon for software preservation:

At least the software code will be preserved in an archive

Security

Measures or provisions in place to ensure software IT security:

The IT department of our institute will be in charge of the deployment

Measures to minimise risks in relation to software development:

The code will be preserved in a GitHub repository, and reviewed by another software developer

Quality Assurance

Governance and Defined Processes

Governance model for software development:

The project leader assigns project responsibilities for each member

Coding standard and quality control: PEP 8 – Style Guide for Python Code

Documentation

Software documentation:

Developers can access the source code in GitHub
Users will be provided with a user manual

Documentation storage and language:

Both documentations, for developers and users, will be stored in a common repository in GitHub

Testing

Test strategies:

- Integration Tests
- Functional Tests
- Tests by end users

Test documentation:

OpenProject

Release and Publish

Releasing

Release process:

Every update will be tested before release

Decision process for releasing and releasing frequency:

Releases will be made if major new features are ready or in case of critical bugs

Software storage and preservation policy:

Source code on GitHub, major releases will be also published in zenodo

Publicly Availability

The software will be publicly available.

Software repository or archive, findability and usability:

GitHub and zenodo links will be published in the institute's website, and the project will be presented in different scientific conferences

Software contributions from the community:

Yes, for example with pull requests in GitHub

Planned (Open) Peer Review for the software:

After the first final version the software will be published in the Journal of Open Source Software

Metadata

Software's metadata assignment:

We will use a software called RDMO, which automatically generates metadata files like readme, codemeta, citation, and license

Support

There will be support and help to re-users of the software.

Support and feedback process with users:

support email address, and feedback also via GitHub

Relation of this Software Management Plan to other Software or Data Management Plan(s):

Yes, for the data measured with our AWS at the institute XXX a DMP exists: Weather Analyser - DMP, which will be available soon

This Software Management Plan will be publicly available in GitHub.

Legal and Ethics

Intellectual Property Rights

Legal ownership of the software:

There is an explicit document describing the owneship of the software

Third-party intellectual or industrial property rights:

No

License

The software will be published under these licenses:

- LGPL-3.0-only
- MIT

Dual Use

Software's use for military purposes:

No